1.9 Composition of Functions

When a car driver depresses the accelerator pedal, it controls the flow of fuel which in turn influences the speed of the car. Likewise, the composition of two functions is a kind of 'chain reaction', where the functions act upon one after another (Fig.1.40).

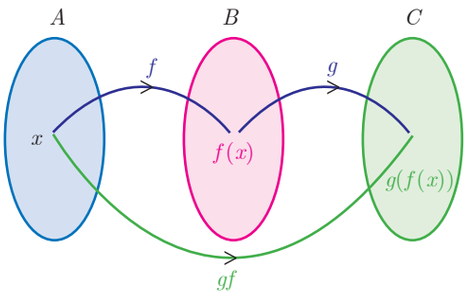


Fig. 1.40

We can explain this further with the concept that a function is a 'process'. If f and g are two functions then the composition $g(f(x))$ (Fig.1.41) is formed in two steps.

(i) Feed an input (say x) to f;

(ii) Feed the output $f(x)$ to g to get $g(f(x))$ and call it $gf(x)$.

Illustration

Consider the set A of all students, who appeared in class X of Board Examination. Each student appearing in the Board Examination is assigned a roll number. In order to have confidentiality, the Board arranges to deface the roll number of each student and assigns a code number to each roll number.

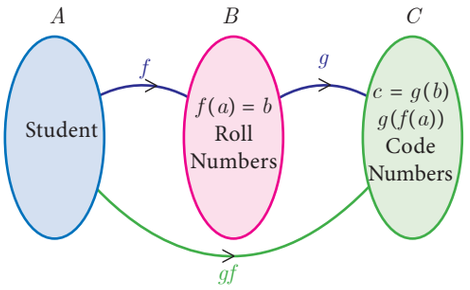


Fig. 1.41

Let A be the set of all students appearing for the board exam. $B \subseteq \mathbb{N}$ be the set all roll numbers and $C \subseteq \mathbb{N}$ be the set of all code numbers (Fig.1.41). This gives rise to two functions $f : A \to B$ and $g : B \to C$ given by $b = f(a)$ be the roll number assigned to student a, $c = g(b)$ be the code number assigned to roll number b, where $a \in A$, $b \in B$ and $c \in C$.

We can write $c = g(b) = g(f(a))$.

Thus, by the combination of these two functions, each student is eventually attached a code number. This idea leads to the following definition.

Definition

Let $f : A \to B$ and $g : B \to C$ be two functions (Fig.1.42). Then the composition of f and g denoted by $g \circ f$ is defined as the function $g \circ f(x) = g(f(x))$ $\forall\ x \in A$.

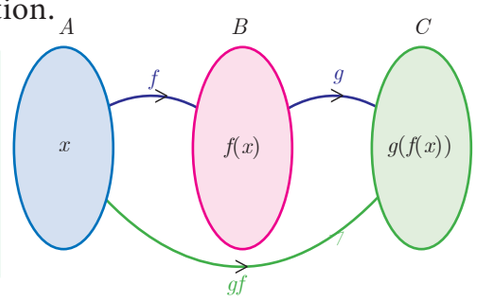


Fig. 1.42

Example 1.19 Find $f \circ g$ and $g \circ f$ when $f(x) = 2x + 1$ and $g(x) = x^2 - 2$

Solution $f(x) = 2x + 1$, $g(x) = x^2 - 2$

$$f \circ g(x) = f(g(x)) = f(x^2 - 2) = 2(x^2 - 2) + 1 = 2x^2 - 3$$

$$g \circ f(x) = g(f(x)) = g(2x + 1) = (2x + 1)^2 - 2 = 4x^2 + 4x - 1$$

Thus $f \circ g = 2x^2 - 3$, $g \circ f = 4x^2 + 4x - 1$. From the above, we see that $f \circ g \neq g \circ f$.

Thinking Corner

If $f(x) = x^m$ and $g(x) = x^n$ does $f \circ g = g \circ f$?

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